MIL-S-36642C (DSA-DM)
18 September 1975
SUPERSEDING
MIL-S-36642B
25 February 1974

MILITARY SPECIFICATION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electrically heated steam sterilizer, suitable for the sterilization of dental and surgical instruments, dressings and flasked solutions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-35	Passivation Treatments for Austenitic, Ferritic and Martensitic Corrosion-Resisting Steel.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-M-7298	Manuals, Technical: Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible.

FSC 6530

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STANDARDS

Federal

FED-STD-66 Steel, Chemical Composition and Hardenability.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11 American Standard Outlet Attachment Plug Caps and Appliance Plugs.

(Copies of ANSI Standards may be obtained upon application to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN WELDING SOCIETY, INC.

Welders Handbook, 6th Edition

(Copies of the Welders Handbook may be obtained upon application to the American Welding Society, Inc., 345 East 47th Street, New York, NY 10017.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 544 STANDARDS FOR SAFETY, MEDICAL AND DENTAL EQUIPMENT.

(Copies of Underwriters' Laboratories Standards may be obtained upon application to the Underwriters' Laboratories, Inc., 207 E. Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Blvd., Santa Clara, CA 95050.)

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U. S. PHARMACOPEIAL CONVENTION, INC. (U.S.P.)

Pharmacopeia of the United States

(Copies of the U.S. Pharmacopeia may be obtained upon application to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof prior to delivery that the items to be delivered under the contract conform to the applicable requirements of UL 544. Proof shall consist of one of the following:

A. UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material. The sterilizer, except for knobs, handles, valves, tubing, controls and insulation, shall be constructed throughout of AISI 300 series corrosion-resisting steel conforming to FED-STD-66 except as specified below. The base plate shall be AISI 300 series corrosion-resisting steel conforming to FED-STD-66. Base plate shall be .062 inch thick minimum. Nuts, bolts, screws, washers and other hardware shall be made of corrosion-resisting steel except hardware not coming into contact with steam or moisture may be of corrosion-resisting steel, nickel-plated steel or cadmium-plated steel. Anodized aluminum for the reset rod and bracket supports shall be acceptable. Lockwashers shall be used wherever machine screws or bolts are utilized for attachment. As an alternate the door may be forged brass. One of the following combinations shall apply for the cabinet wraparound, front panel and back panel.

(a) 0.035 inch thick (No. 21 United States Standard (U.S.S.) gage) minimum AISI 302 corrosion-resisting steel for all three components.

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(b) 0.035 inch thick (No. 21 New Birmingham gage) minimum, carbon steel for the wraparound and back panel; front panel shall be 0.044 inch thick (No. 19 New Birmingham gage) minimum, carbon steel.

(c) 0.017 inch thick (No. 27 U.S.S. gage) AISI 430 corrosion-resisting steel with a stamped pattern for rigidity on the wraparound and back panel; front panel shall be 0.050 inch thick (No. 18 U.S.S. gage) AISI 430 corrosion-resisting steel. The stamped pattern shall be a geometrical figure repeated continuously and uniformly over the entire area of the sheet metal.

(d) 0.031 inch thick (No. 22 U.S.S. gage) wraparound, 0.025 inch thick (No. 24 U.S.S. gage) rear panel, and 0.062 inch thick (No. 16 U.S.S. gage) front panel; all three pieces AISI 302 corrosion-resisting steel.

3.3 Design and construction. Shall be a portable, electrically-heated, automatically or manually controlled, single shell, steam sterilizer encased in a cabinet. The unit shall consist of a sterilizing chamber, combination water reservoir and steam condensing tank, electric heater, gages, controls, valves and trays. The sterilizer shall operate on 115 volts, 60 Hertz, alternating current (AC).

3.3.1 Cabinet. The cabinet material shall be as specified in 3.2. Maximum dimensions of the cabinet shall be 15 inches wide by 17-3/4 inches deep by 15-1/4 inches high. The cabinet shall be secured to a base plate by not less than six cadmium plated or corrosion-resisting steel screws. The base plate shall be dimensioned to accept the cabinet shell. The front panel shall be firmly secured to the base plate. A neoprene rubber suction cup or rubber tipped leg shall be located at each corner of the base.

3.3.2 Sterilizing chamber. The corrosion-resisting steel chamber shall be seamless or all-welded construction and shall be 14-1/2 inches ± 5/8 inch long by 7-1/2 inches ± 5/8 inch in diameter. The entire chamber shall be passivated in accordance with QQ-P-35. Except for the area near the operating thermostat and low-water cutoff thermostat, the entire chamber shall be covered with 1 inch thick glass fiber insulation having a foil face on the outside. The glass fiber insulation shall be retained by four 0.006 inch thick minimum by 3/4 inch wide aluminum straps; two around the chamber circumference and two crossed at the end of the chamber. The straps shall be fastened in such a manner as to cause only slight compression of the glass fiber and thus provide maximum insulation. The chamber shall be furnished with one small and two large instrument trays or three large trays. The bottom of the trays shall be perforated over a minimum area of 3 inches by 10-1/2 inches for the small tray and 4-1/2 inches by 10-1/2 inches for the large trays. A rack insert or other suitable means shall be provided to support the trays, one above the other with a minimum 1/2 inch between the top of one tray and the bottom of the tray immediately above. A fill indicator shall be provided to indicate proper level of water to be added to the chamber.

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3.3.2.1 Chamber door. The chamber door shall be constructed of corrosion-resisting steel or forged brass. The chamber door shall contain a silicone rubber gasket for sealing the chamber when the door is closed. The chamber door shall be capable of one-hand operation easily inserted, locked, unlocked, and opened without undue pressure or force. Gaskets which have cemented ends shall withstand the test specified in 4.4.16 without separating. If special technique or maintenance is required for proper operation of door, permanent, legible instructions shall be affixed in a location which is readily visible to the operator.

3.3.3 Reservoir. A welded, not brazed or soldered, corrosion-resisting steel reservoir shall be located within the cabinet. The reservoir shall contain not less than 3-1/2 quarts of water when filled to a clearly visible fill marking. The fill opening shall be located at the top surface of the cabinet and shall be provided with a corrosion-resisting steel or phenolic cap. The reservoir cover shall seal the tank sufficiently to prevent spilling when the sterilizer is tilted 30 degrees in any direction when tested as specified in 4.4.15. There shall be no escape of water during the venting operation as specified in 4.4.5. A copper or brass tubing coil shall be located at the bottom of the reservoir to condense steam on its return to the reservoir after sterilization cycle. Through the wall fittings for plumbing fixtures shall be brazed or welded, leaktight, to the reservoir. Plumbing fixtures requiring maintenance shall be removeable. The reservoir seams shall be passivated in accordance with QQ-P-35.

3.3.3.1 Draincock. A draincock for emptying the reservoir shall be located on the bottom of the sterilizer. There shall be sufficient clearance under the unit so that the reservoir drain will not touch the surface on which the sterilizer is resting.

3.3.4 Electric heater. An electric heating element of the immersion type, with a power consumption rating of not more than 1450 watts, shall be provided. As an alternate, three resistance wire and mica heating elements shall be strapped securely to the outside of the chamber. The heater shall be capable of producing sufficient steam to provide required performance (see 3.4).

3.3.5 Heater cutoff. There shall be a low-water cutoff for the immersion-type heater, or an overheat protector when the heating elements are external to the chamber.

3.3.5.1 Low-water cutoff. When immersion heaters are used, a low-water cutoff shall protect the heater against burnout when the water level in the chamber drops to expose the heating elements. The cutoff shall operate to cut the flow of current to the heater within 7 minutes. The cutoff shall have a manual reset. When the cutoff has operated, it shall not cycle to the "on" position when tested as specified in 4.4.7. There shall be positive indication that the low-water cutoff has functioned. The low-water button shall not be recessed within the front panel when pressed to the "reset" position. The word "Reset" shall identify the button.

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3.3.5.2 Overheat protector. The overheat protector shall be a thermostat located at the heating elements under the sterilizing chamber and shall be connected to the center element only. The thermostat shall protect the chamber from overheating should the temperature-pressure control fail, or the sterilizer be set to sterilize without water in the chamber and the temperature-pressure control fails to sense heat rapidly enough. The overheat protector shall turn off power to the heating elements when tested as specified in Test A of 4.4.8, and shall not function when tested as specified in Test B.

3.3.6 Valves. Valves shall be provided as specified herein.

3.3.6.1 Safety relief valve. Each sterilizer shall be equipped with a safety valve which shall relieve the pressure in the chamber by venting into the reservoir at 33 pounds per square inch gage (p.s.i.g.) ± 3 p.s.i.g. and which shall reseal within 5 p.s.i.g. below the cracking pressure. The safety valve shall be tested on the sterilizer for the aforementioned requirements in accordance with 4.4.9.

3.3.6.2 Filler-exhaust valve. A manually-operated filler-exhaust valve shall be provided to control filling of the chamber to the desired level. This same valve shall also allow the return of the residual steam and condensate from the chamber to the reservoir in the exhaust cycle. Materials used for filler-exhaust valve shall be suitable for the high temperature application as specified in 4.4.14.

3.3.7 Temperature gage. A dial-type temperature gage, graduated over a minimum range of 210° F. to 270° F., shall be provided. The temperature gage shall be accurate to within ± 2° F. at the 250° F. and 270° F. settings when tested as specified in 4.4.10. The temperature gage sensing element shall be located in the discharge line or in the same line as the safety valve. The thermometer reading shall be within +0° F. and -4° F. when tested as specified in 4.4.12. The gage shall be securely attached to the back of the front panel.

3.3.8 Indicator light. A green indicator light on the front panel shall indicate that the heater is in operation.

3.3.9 Timer switch and audible alarm. A variable timer switch with a maximum setting of not less than 30 minutes, shall be provided to start the heater and to turn it off at the end of the sterilizing period. As an alternate, a separate switch for starting and turning off the heater, and a separate timer, shall be acceptable. The timer shall activate an audible signalling device when the selected sterilization cycle has been completed. A zero reference mark for the timer shall be provided on the front panel. The timer shall be accurate to plus 2, minus zero minutes at the 30 minute setting when tested as specified in 4.4.11. The timer shall be so designed and constructed that tightening of the mounting screws will not cause interference with the timer operation.

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3.3.10 Temperature (pressure) control. The sterilizer shall be equipped with a temperature (pressure) control so arranged that the temperature (pressure) shall be attained and held as described in 3.4.

3.3.11 Line cord and plug. Each sterilizer shall be equipped with a line cord not less than 6 feet in length. The cord shall be HSJ, SJT, or SJO three-wire type with one ground lead and shall be fitted with a three-prong plug with grounding blade capable of fitting into a receptacle specified in the American National Standards Institute (ANSI) Standard C73.11. Internal wiring shall be harnessed to prevent contact with hot portions of the sterilizer. If type HSJ cord is used, asbestos insulation on the wiring shall be evenly placed and closely tied where the outer jacket of the cord is removed for connection of the wiring to the sterilizer. Polarity of the line cord shall be in accordance with UL standards of safety.

3.3.12 Repair parts. The following repair parts shall be provided with each sterilizer:

(a) One heating element assembly complete with gaskets and terminal hardware for the immersion type, or one heating element assembly complete with terminal hardware for the external type.

(b) One gasket (chamber or door, as applicable).

(c) One of each lamp.

(d) Where required for proper door operation, one tube containing at least two ounces of gasket lubricant.

3.4 Performance. The sterilizer shall be capable of at least ten complete sterilization cycles on one filling of the reservoir when tested as specified in 4.4.13. There shall be no leakage of steam nor of water around the chamber door from the time power is applied to the sterilizer. Slight wisps of steam around the door up to a temperature (pressure) gage reading of 220°F. shall not be considered steam leakage. The sterilizer shall be provided with means to allow air evacuation from the chamber thermostatically until the chamber is filled with steam, at which time all vents shall close automatically. Slight wisps of steam from the steam trap shall be acceptable, providing the set temperature holds. The automatic temperature (pressure) switch shall operate when the temperature (pressure) in the chamber reaches the desired values and shall cycle to maintain that pressure until the sterilizing time is complete. The temperature (pressure) control shall hold the selected temperature, either 250° F. or 270° F., to +2° F., -0° F. for 10 consecutive minutes when tested as specified in 4.4.4. Starting with water at 75° F., the sterilizer shall be capable of attaining

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spore killing temperatures and steam pressures (250°F. and 15 to 17 p.s.i.g.) within 10 minutes, and (270° F. and 27 to 30 p.s.i.g.) within 11 minutes, when tested as specified in 4.4.6. The sterilizer shall be capable of killing 100 percent of spores on test strips (see 6.6) when tested as specified in 4.4.6. The sterilizer shall pass the operation test specified in 4.4.5. The door shall be incapable of being opened while there is steam pressure in excess of 3 p.s.i.g. in the chamber as indicated on the temperature (pressure) gage or when the thermometer reads above 212° F., whichever applies. All controls and gages shall be readily accessible. The sterilizer shall be capable of being operated without danger of scalding to the operating personnel. Steam shall be exhausted into the reservoir upon completion of the sterilization process. There shall be no electrical breakdown when the sterilizer is subjected to the dielectric strength test specified in 4.4.2.

3.4.1 Temperature of various parts. The temperature of the following parts shall not exceed the maximum temperatures specified in UL 544 (see 4.4.1):

Wire nuts.
Jacket of cord near strain relief.
Wire near jacket of sterilizer.
Plastic lens of pilot light.
Center of supporting surface.
Rubber foot support.
Wire near heater terminal.
Areas which are likely to be touched by operator during use.

3.4.2 Pressure cylinder (sterilizer chamber assembly) hydrostatic strength. The pressure cylinder sterilizer chamber assembly shall withstand a pressure of not less than 180 p.s.i.g. for a period of one minute when tested as specified in 4.4.3. The sterilizer shall be so constructed that there shall be no leakage or sweating, except around the door, and no permanent set as a result of the hydrostatic pressure test.

3.4.3 Endurance. The sterilizer shall pass the endurance test specified in 4.4.14.

3.5 Finish. The corrosion-resisting steel sterilizer cabinets and front panels shall have No. 3 or better commercial finish. Carbon steel cabinets and front panels shall be painted with a baked-on, heat-resistant paint on the exterior and primed and painted on the inside to prevent corrosion. Paint shall be applied uniformly and shall be free of runs, chips, cracks, blisters and orangepeel. If the chamber door is forged brass it shall be chrome over nickel plated.

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3.6 Marking.

3.6.1 Instruction plates. Metal instruction plates or metal-foil decals bearing detailed instructions for operation of the sterilizer; a warning, if an electrically operated timer is used, that sterilizer is to be operated on alternating current only; and a statement requiring replenishment of the water supply, shall be permanently attached to the sterilizer and so located as to be readily visible or available to the operator. Marking shall be clear, legible, and permanent. Artwork and wording for the instruction plate shall be submitted to the Contracting Officer for approval when a First Article sample is required, unless prior approval has been received.

3.6.2 Identification marking. Each sterilizer shall also have, permanently attached by spot welding or rivetting, a nameplate marked with the following information:

National Stock Number
Serial number
Manufacturer's name or registered trademark
Part or model number
Contract or purchase order number
Letters U. S.
Electrical rating (amperes, volts, hertz and wattage)

As an alternate, the above information divided between two nameplates shall be acceptable. It shall be possible to identify the unit with the cover removed thus preventing loss of original item identification during maintenance.

3.7 Service data. The contractor shall furnish, with each unit, two copies of a service data manual conforming with MIL-M-7298. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. This service data shall contain complete instructions for installation, operation (including a recommended time and temperature sterilization table), maintenance, and a list of component parts. The service data shall include sufficiently detailed information to permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review

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and comment by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 addiitonal copies may be waived upon request to the contracting officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.7.1 Instructions for shipboard installations. The service data shall include the following instructions for shipboard installation:

A. If at all practicable, the sterilizer shall be so installed that the chamber axis is parallel to the ship's centerline, rather than athwartship, to minimize the effect of the ship's roll.

B. Installation of the sterilizer shall increase the original backward slope of its chamber, as follows:

1. Remove the tray rack.

2. Install a 1/8 inch IPT brass street ell inside the chamber, on the male threads of the air vent connection in the lower portion of the chamber backhead. When lightly tightened with a wrench or pliers, the male-threaded outlet of the street ell should point vertically upward; if not, sand, file or grind the flat surface of the female-threaded end of the street ell as required for proper orientation. The street ell is required to prevent water in the chamber from draining through the air vent opening as the ship pitches.

3. Measure the original backward slope of the chamber (with the sterilizer horizontal), by such means as the protractor head of a machinist's combination square, placed in a vertical plane across the upper and lower edges of the chamber mouth, with the door open. Raise the front of the sterilizer on suitable mounting blocks or spacers, to increase the chamber's backward slope by $5^{0} + 1^{0}$. For example, if the original slope (with the sterilizer horizontal) is 3^{0}, this angle should be increased to $8^{0} \pm 1^{0}$.

4. Pour water directly into the chamber until the highest point of the electric heating elements is submerged 3/8 inch below the water surface.

5. Replace the tray rack.

6. Mark the new filling level on the rack, at the height now reached by the water.

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7. Revise the filling instructions in the sterilizer service data accordingly.

C. A suitable receptacle shall be used for containing any unavoidable residual water spillage, when the door is opened after completion of a sterilization cycle. This shall take the form of a tray or similar container, whose rear face shall be inserted under the sterilizer front before the chamber door is opened, and whose forward face shall provide a suitable baffle.

D. The operator shall stand clear of the possible path of water spillage, when opening the sterilizer door.

3.8 Workmanship. Workmanship shall be first class throughout. Sterilizers shall be free from defects which detract from their appearance or may impair their serviceability. All burrs and sharp edges shall be removed from the sterilizer.

3.8.1 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal and/or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Classification of inspection. Quality conformance inspection consists of examination and tests, performed on the sterilizer manufactured and submitted for acceptance under contract, to determine that the sterilizer submitted for acceptance under the contract is in conformance with the specification.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

TABLE I

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0
For dimensional examination	S-3	6.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. The acceptable level shall be zero.

TABLE II

Component	Characteristic	Requirement	Test procedure	Inspection level
Sterilizer	Hydrostatic pressure	3.4.2	4.4.3	1/
	Low-water cutoff	3.3.5.1	4.4.7	S-3
	Overheat protector	3.3.5.2	4.4.8	S-3
Safety relief Valve	Accuracy	3.3.6.1	4.4.9	100%
Temperature gage	Accuracy	3.3.7	4.4.10	100%
Sterilizer	Dielectric strength	3.4	4.4.2	100%
	Spore kill	3.4	4.4.6	S-2
	Operation	3.4	4.4.5	100%
Sterilizer	Temperature of various parts	3.4.1	4.4.1	1/
Temperature (pressure) control	Accuracy	3.3.10	4.4.4	S-1
Sterilizer	Chamber temperature	3.3.7	4.4.12	S-1
Sterilizer	Cycles per fill	3.4	4.4.13	S-1
Sterilizer	Endurance	3.4.3	4.4.14	1/
Timer	Accuracy	3.3.9	4.4.11	100%
Reservoir	Spillage	3.3.3	4.4.15	S-1
Gasket	Pull Test	3.3.2.1	4.4.16	S-2

1/ One per contract - change in component design or manufacturer requires additional test.

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4.3 Examination. The sterilizers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects in Table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III

Major	
101	Instruction plate missing, not permanently secured to sterilizer, incomplete, or illegible.
102	Sterilizer does not function at specified temperatures.
103	Any component missing, broken, or otherwise unusable.
104	Service data missing or does not include all specified information.
105	Exposed edges not free of burrs and sharpness.
106	Door does not open and close freely without binding.
107	Temperature gage does not have minimum range specified.
108	Low-water cutoff reset button does not extend beyond front of sterilizer in cutoff condition.
109	Low-water cutoff reset button recessed within front of sterilizer in reset condition.
110	Line cord not free of reverse polarity.
111	Electrical connections not tight and/or secure.
112	Plumbing not properly routed, not interchangeable from unit to unit, not free of kinks or flattened areas.
113	Electrical terminals not the correct size.
114	Welding not continuous, not free of voids, holes or cracks.
115	Fill valve does not operate smoothly without binding or sticking.
Minor	
201	Exposed corrosion-resisting steel surfaces not No. 3 finish or better.
202	Finish not free of scratches.
203	Sterilizer does not rest evenly on flat surface.
204	Safety valve not marked with discharge pressure value.
205	Condensing coil not free of kinks and sharp bends.
206	Nameplate missing, incomplete, illegible.
207	Plumbing connections not properly made, not cleanly dressed, not first class throughout.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Component accuracy (testing as specified in 4.4.10 through 4.4.11) may be performed at any time prior to installation on the sterilizer provided test reports are retained and submitted with documentation applicable to each sterilizer.

4.4.1 Temperature of various parts. The sterilizer shall be operated at the rated or adjusted voltage as described in UL 544, with the sterilizing chamber filled with water, until constant temperatures are reached. These temperatures and the ambient at which the sterilizer is tested shall be reported.

4.4.2 Dielectric strength. While in a heated condition, following the temperature test in 4.4.1, the sterilizer shall be subjected to a gradually increasing 60 Hertz potential of 1000 volts, plus twice the rated voltage, applied between the current-carrying parts and the metal enclosure and maintained at the maximum value for one minute.

4.4.3 Hydrostatic pressure. The pressure cylinder (sterilizing chamber) shall be connected by suitable fittings and tubing to a hydrostatic pressure source. The cyliner shall be gradually filled with water and subjected to the pressure requirement of 3.4.2.

4.4.4 Accuracy of temperature (pressure) switch. A calibrated maximum-reading thermometer shall be inserted in the center of the sterilizer chamber and the temperature recorded at the 250°F., 15 p.s.i.g. setting and at the 270°F., 27 p.s.i.g. setting.

4.4.5 Operation. Each sterilizer shall be operated in accordance with the manufacturer's instructions to determine compliance with requirements herein. The following shall be observed as to whether:

(a) Timer switch or separate heater switch, whichever applies, activates heater and indicator light.
(b) Thermostatic air vent allows all air to be evacuated from chamber.
(c) Door cannot be opened at any pressure in excess of 3 p.s.i.g. any temperature in excess of 212°F., whichever applies.
(d) Temperature gage registers.
(e) Temperature switch cycles to maintain temperature (pressure).
(f) Time deactivates heater and lights.
(g) Time to reach 250°F. and 15 p.s.i.g., and 270°F. and 27 p.s.i.g., is in compliance with this specification.
(h) Door effects a positive seal (see 3.4).
(i) Water escapes from reservoir when chamber is vented (see 3.3.3).

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4.4.6 Temperature. The sterilizer shall be operated at 250°F. + 2° - 0°F. for 15 minutes and at 270°F. + 2° - 0° F, for 3 minutes as determined by a calibrated maximum reading thermometer in the center of the chamber, and in accordance with the manufacturer's instructions. Two sterilization test strips of the type described in 6.6, each wrapped in a towel, shall be used to determine sterilization efficiency at the 250° F. and 270° F. settings. A spore strip from the same lot is to be retained as a viability control. Testing for spore kill shall be as recommended in the U. S. Pharmacopeia.

4.4.7 Low-water cutoff. With no water in the sterilizer, the timer shall be set for 30 minutes. Note when the cutoff has functioned.

4.4.8 Overheat protector. The overheat protector shall meet the performance requirements of 3.3.5.2 for the following applicable tests:

Test A. The sterilizer shall be set at the Sterilize position and run without water in the chamber.

Test B. From a cold start, the sterilizer shall be run up to 270°F.

4.4.9 Safety relief valve. The sterilizer shall be filled with water to the fill level mark and the temperature/pressure regulator bypassed to prevent it from cutting off the power to the heating elements. The pressures at which the valve relieves and reseals shall be recorded.

4.4.10 Accuracy of temperature gage. The accuracy of the temperature gage shall be determined using equipment accurate to the nearest 0.5°F.

4.4.11 Accuracy of timer. The timer shall be tested against a stopwatch and the accuracy recorded.

4.4.12 Chamber temperature reading. Chamber temperature shall be measured by means of a themocouple or calibrated maximum-reading thermometer at the midpoint of the chamber during a sterilization cycle after the temperature has stabilized.

4.4.13 Sterilization cycles per filling. Ten consecutive and complete sterilization cycles shall be run with the chamber temperature maintained at 270°F. + 2°, -0°F. for three minutes in each cycle.

4.4.14 Endurance. The test described in 4.4.13 shall be performed, but extended to 100 complete sterilization cycles.

4.4.15 Reservoir spillage. The sterilizer shall be placed on a level surface and the reservoir filled with water to its fill line. The sterilizer then shall be tilted to 30° fore, aft, left and right. Any spillage at each of the four postures shall be reported.

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4.4.16 Gasket. A pull of 10 pounds shall be exerted across the cemented area of this type gasket.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. Each sterilizer, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method III.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be packed in an exterior container designed for type 3 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, type CF, class weather-resistant. Fiberboard boxes shall conform to the special requirements specified in Table II of PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be packed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. In addition, special requirements specified in Table I of PPP-B-636 shall apply for fiberboard boxes.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery called for in the contract or purchase order.

5.3 Marking. Marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. The sterilizer described herein is intended for use in the autoclave sterilization of dental and surgical instruments, and dressings and flasked solutions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the product covered by this specification. The activity responsible for the Qualified Products List is the Defense Personnel Support Center, Directorate of Medical Materiel, 2800 South 20th Street, Philadelphia, PA 19101 and information pertaining to qualification of products may be obtained from that activity.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover types which are normally procured to meet requirements.

6.5 This specification covers the following item shown in the Federal Supply Catalog:

NSN	Item Identification
6530-00-782-6503	STERILIZER, SURGICAL INSTRUMENT AND DRESSING, Pressure, Electrically Heated, Corrosion-Resisting Metal, 7-1/2 by 14-1/2 inches, Automatic Control, 115 volt, 60 Hz, AC

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6.6 The sterilization test strips referenced in 4.4.6 are available as Castle Company's Unispore and American Sterilizer Company's Spordi and are as shown in the Federal Supply Catalog:

NSN	Item Identification
6530-00-477-6720	Sterilization Test Strips Set, Bacterial Spore.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6530-1375
is

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-36642C(DSA-DM)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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